

Comparing Attack Models for IT Systems: Lockheed Martin's Cyber Kill Chain, MITRE ATT&CK Framework and Diamond Model

Nitin Naik¹, Paul Jenkins², Paul Grace¹ and Jingping Song³

¹School of Informatics and Digital Engineering, Aston University, United Kingdom

²Cardiff School of Technologies, Cardiff Metropolitan University, United Kingdom

³Software College, Northeastern University, China

Email: n.naik1@aston.ac.uk, pjenkins2@cardiffmet.ac.uk, p.grace@aston.ac.uk and songjp@swc.neu.edu.cn

Abstract—Cyberattacks are a serious challenge for any IT system, which motivates cybersecurity professionals and developers to constantly explore various attack models to analyse the cyber-attack lifecycle used by attackers to mount their attacks. Several attack models have been proposed and successfully utilised to analyse the various types of cyberattacks and their mitigations. All the models offer different characteristics, advantages and disadvantages. Therefore, it is important to examine the most popular attack models to understand their characteristics and suitability for modelling specific types of attacks. This paper will examine characteristics, advantages and disadvantages of the three most popular attack models: the Lockheed Martin's Cyber Kill Chain, the MITRE ATT&CK Framework and the Diamond Model. Finally, it will provide a comparative study of these three attack models to present a relative summary for selecting the most suitable attack model for a specific application.

Index Terms—Cyberattack; Attack Models; Lockheed Martin's Cyber Kill Chain; MITRE ATT&CK Framework; Diamond Model.

I. INTRODUCTION

Cyberattacks are ever evolving and constitute the biggest challenge for the security of IT systems, requiring cybersecurity professionals and developers to stay ahead of this challenge and minimise the risk of cyberattacks [1], [2]. Attack models are one of the most effective tools to manage this challenge and are used to identify, document and simulate attacks against IT systems encompassing various attack techniques and paths and therefore the effective mitigations of these attacks [3]. Using these attack models, cybersecurity professionals and developers better understand the behaviour, tactics and objectives of attackers and can take appropriate steps to remediate any vulnerabilities within their IT systems. The aforementioned benefits of attack models contribute to ensuring it is one of the most successful security tools in the security industry, and cybersecurity professionals and developers constantly explore various attack models for their benefits and suitability to analyse the cyberattack lifecycle used by attackers to mount their attacks.

Several attack models have been proposed and successfully utilised to analyse various types of cyberattacks and their mitigations [4], [5], [6] [7], [8], [9]. All the models

offer different characteristics, advantages and disadvantages. Therefore, it is important to examine the most popular attack models to understand their characteristics and suitability for modelling specific types of attacks. This paper will examine the characteristics, advantages and disadvantages of the three popular attack models: the Lockheed Martin's Cyber Kill Chain, the MITRE ATT&CK Framework and the Diamond Model. The Lockheed Martin's Cyber Kill Chain (CKC) is a phase-based attack model to assist security experts to understand the constituent parts of an attack consisting of seven different steps and develop a defence by disrupting any of one or more of these attack steps [4]. The MITRE ATT&CK Framework is a globally accessible knowledge base of adversarial tactics and techniques based on real-world observations, reflecting the various phases of an adversary's attack lifecycle and the platforms they are known to the target [5]. The Diamond Model utilises the Diamond structure to emphasize the relationships and characteristics of an attack based on its four core components: adversary, infrastructure, capability, and victim [6]. Finally, it will provide a comparative study of these three models to present a relative summary for selecting the most suitable attack model for a specific application.

This paper is structured in the following sections: Section II will examine characteristics, advantages and disadvantages of the Lockheed Martin's Cyber Kill Chain (CKC). Section III will examine characteristics, advantages and disadvantages of the MITRE ATT&CK Framework. Section IV will examine characteristics, advantages and disadvantages of the Diamond Model. Section V presents a comparative study of the three attack models: the Lockheed Martin's Cyber Kill Chain, the MITRE ATT&CK Framework and the Diamond Model. Section VI summarises the paper and outlines some future work.

II. LOCKHEED MARTIN'S CYBER KILL CHAIN (CKC)

The Lockheed Martin's Cyber Kill Chain (CKC) is a phase-based attack model to assist security experts understand the breakdown of an attack into seven different steps and develop

its defence by disrupting any one of these attack steps [4]. It identifies the following sequence of attack steps: Reconnaissance, weaponization, delivery, exploitation, installation, command and control, and action on objectives (see Fig. 1). It is developed by Lockheed-Martin, which is co-opted from the military term kill-chain used to break down the structure of an attack [10]. The CKC concept is that a defender is required to only disrupt one attacking step in the chain to stop that attack.

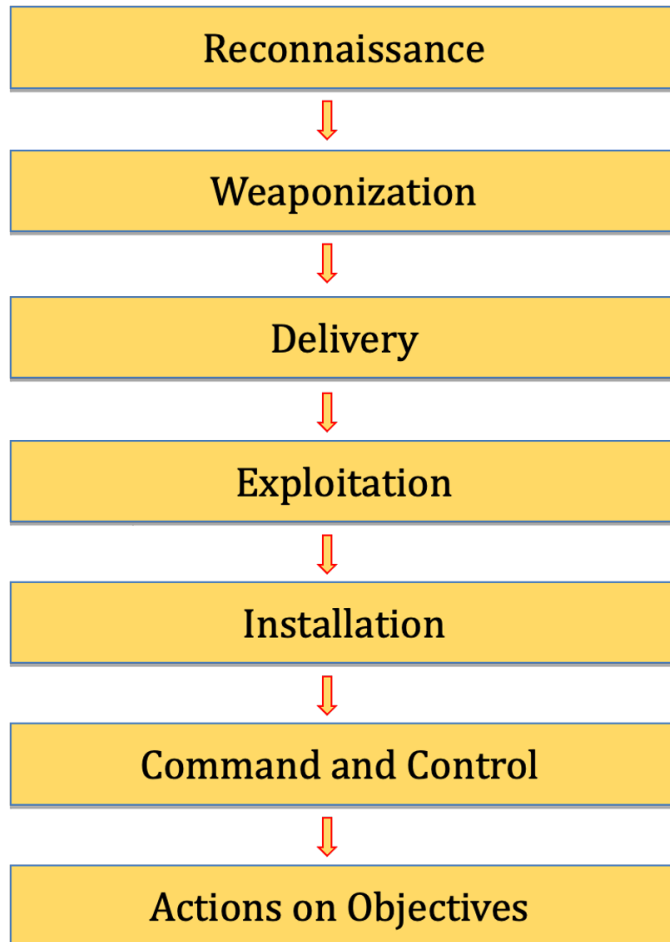


Fig. 1. Steps of Lockheed Martin's Cyber Kill Chain (CKC) of Intrusion Analysis

A. Steps of Lockheed Martin's Cyber Kill Chain

The Lockheed Martin's Cyber Kill Chain contains seven steps as shown in Fig. 1, and are described here:

1) *Reconnaissance*: In this step, an attacker identifies both targets and tactics for the attack and conducts an extensive investigation on the target to identify its vulnerabilities that can be exploited. An attacker gathers all the necessary information relating to the target either using manual methods or automated scanners designed for mapping organisational structures, to identify any security gaps and vulnerabilities which can be exploited.

2) *Weaponization*: In this step, an attacker utilises the information gathered during the reconnaissance phase to create the cyber weapon such as malware, DDoS, and botnet to exploit the found vulnerabilities. Depending on the target and the purpose of the attacker, this cyber weapon can also exploit new, undetected vulnerabilities or a combination of different vulnerabilities.

3) *Delivery*: In this step, an attacker transmits the cyber weapon to the target for the purpose of exploiting the found vulnerabilities. An attacker can use any transmission method to transmit cyber weapons such as emails and websites.

4) *Exploitation*: In this step, an attacker starts exploiting the target's vulnerabilities using the transmitted weapon to attempt to gain wider access in the target network.

5) *Installation*: In this step, an attacker installs additional programs such as a web shell on a compromised web server or a backdoor implant on a compromised system, enabling the attacker to remain undetected, bypass security controls, escalate privileges, and extract password hashes.

6) *Command and Control*: In this step, an attacker establishes a Command and Control (C2), channel to operate and monitor an attack remotely. Command and Control of a compromised system is usually accomplished through a beacon (malicious payload), which sends regular communications from a compromised system to an attacker-controlled system to communicate and request instructions.

7) *Actions on Objective*: In this step, an attacker takes all the necessary actions to achieve the final goal of an attack. The attacker's final goal could be anything from data exfiltration, data modification, data destruction and data encryption.

B. Advantages of Lockheed Martin's Cyber Kill Chain

The Lockheed Martin's Cyber Kill Chain has the following advantages [4], [10], [11]:

- The Cyber Kill Chain (CKC) model is a comprehensive framework for attack analysis based on the Intelligence Driven Defence model. It provides a comprehensive understanding of an attack, an attacker, and attack actions by dissecting each step of the chain for both pre-incident and post-incident analysis.
- The CKC model provides an understanding of defence capabilities and security gaps of an organisation based on the employed threat detection tools and threat actors.
- Based on this comprehensive understanding of the CKC model, IT security teams can determine where to focus their defence mechanisms for the most effective outcome and reducing the risk of experiencing significant disruption or loss.
- The CKC model is one of the most informative models used in focusing on the human element, contrary to several other models which are mainly focused on the technology. This enables organisations to better protect, detect, and respond to cyberattacks based on the human element and utilising it for applying to and managing human risk.

C. Disadvantages of Lockheed Martin's Cyber Kill Chain

The Lockheed Martin's Cyber Kill Chain has the following disadvantages [3], [5], [12], [13] :

- The CKC model applies the century-old military kill chain model to a cyberattack, however, it has several security gaps as it remains mostly unmodified since its creation.
- The CKC model works well to protect against malware and Advanced Persistent Threats (APTs) and does not account for sophisticated and modern methods that attackers currently use to attack an environment. However, the CKC model has evolved in order to anticipate and recognize advanced threats such as ransomware and social engineering-based attacks.
- The CKC model does not account for such iterative approaches or combinations of Tactics, Techniques and Procedures (TTP) that adapt according to the encountered environment.
- The CKC model can detect an attack on its various steps. However, an analysis conducted in 2013, wherein the US senate discovered that the different steps of the protocol could not detect an attacks' progression.
- The CKC model is mainly an attackers process from beginning to end, focusing on the attackers goals, actions, executions and achievements, which renders it as a less flexible approach.
- The CKC model is an excellent model for the defence of external attacks, though, it is not suitable for insider threats. However, the CKC has evolved in order to anticipate and recognize insider threats.
- The timescale of the CKC steps is uneven, meaning that first five steps are smaller than the last two steps.

III. MITRE ATT&CK FRAMEWORK

The MITRE ATT&CK stands for Adversarial Tactics, Techniques, and Common Knowledge, which is a framework created by MITRE in 2013 [14]. This model is a globally accessible knowledge base of adversarial tactics and techniques based on real-world observations, reflecting the various phases of an adversary's attack lifecycle (see Fig. 2) and the platforms they are known to the target [5]. The ATT&CK framework comprises a structured list of known adversary behaviours that have been compiled into tactics and techniques, expressed as a series of matrices [14]. The tactics and techniques abstraction in the model provide a common taxonomy of individual adversary actions understood by both offensive and defensive sides of cybersecurity [10]. Furthermore, it provides an appropriate level of categorization for adversary action and specific mechanisms of defending against it [5]. The ATT&CK knowledge base is used as a foundation for the development of specific threat models and methodologies in the private sector, government, and the cybersecurity product and service community [14].

Reconnaissance
Resource Development
Initial Access
Execution
Persistence
Privilege Escalation
Defence Evasion
Credential Access
Discovery
Lateral Movement
Collection
Command & Control
Exfiltration
Impact

Fig. 2. Various Tactics of MITRE ATT&CK Framework of Intrusion Analysis

A. Tactics and Techniques of MITRE ATT&CK Framework

The MITRE ATT&CK framework is a collection of several intrusion techniques grouped into 14 different tactics [14] as shown in Fig. 2, which are as follows:

- 1) *Reconnaissance*: The attacker is trying to gather information to plan future operations to accomplish an attack.
- 2) *Resource Development*: The attacker is trying to establish resources to support future operations to accomplish an attack.
- 3) *Initial access*: The attacker is trying to gain an initial access to the target system.
- 4) *Execution*: The attacker is trying to run malicious code on the local system or the remote system.
- 5) *Persistence*: The attacker is trying to maintain continuous access to the target system across restarts, changed

credentials, and other interruptions that could terminate their access.

6) *Privilege escalation*: The attacker is trying to gain higher-level permissions and elevate access on the target system.

7) *Defence evasion*: The attacker is trying to avoid detection throughout their compromise of the target system.

8) *Credential access*: The attacker is trying to steal credentials to access the target system.

9) *Discovery*: The attacker is trying to gain more knowledge about the target system and its internal environment.

10) *Lateral movement*: The attacker is trying to enter and move through the internal environment of the target system.

11) *Collection*: The attacker is trying to gather information relevant to the attacker's aim.

12) *Command and Control*: The attacker is trying to communicate with the compromised system to control and operate that system.

13) *Exfiltration*: The attacker is trying to steal data from the compromised system.

14) *Impact*: The attacker is trying to manipulate, interrupt, or destroy the compromised system and its data.

B. Advantages of MITRE ATT&CK Framework

The MITRE ATT&CK Framework has the following advantages [5], [10], [14], [15]:

- The MITRE ATT&CK framework is a globally-accessible knowledge base of adversary tactics and techniques based on real-world observations. This abstraction of tactics and techniques provides a common taxonomy of an attack and attacker actions understood by both offensive and defensive sides of cybersecurity and serves as a centralized vocabulary for industry professionals.
- The MITRE ATT&CK framework offers a detailed response phase, where it maps attack patterns against courses of action, which is moderately beneficial and not available in several well-known attack models.
- The MITRE ATT&CK framework includes various functions and roles across the security team. It aids in the resolution of some of the cyber skills training issues, as junior security analysts can perform some analysis based on the knowledge base and following industry standard procedures and defence strategies.
- The MITRE ATT&CK framework can be used as a basis to develop specific threat models and methodologies in the private sector, government, and the cybersecurity product and service community.
- The MITRE ATT&CK framework is used for a wide range of security analyses such as intrusion detection, threat hunting, security engineering, threat intelligence, red teaming, and risk management.

C. Disadvantages of MITRE ATT&CK Framework

The MITRE ATT&CK Framework has the following disadvantages [3], [5], [16]:

- The MITRE ATT&CK framework provides an appropriate level of classification for adversary actions and specific techniques for defending against it, however, the main issue is that hierarchical structures are missing or inconsistent.
- Not all identifiers of tactics and techniques are traceable, which lacks linearity, grouping and/or hierarchy. This means that these techniques cannot be assigned exclusively to individual tactics and are often used by multiple tactics and across multiple phases of an attack.
- The MITRE ATT&CK framework utilises some generic definitions of attacks, whereby some attacks are considered the same, as is the case of network DoS, which can be caused through bandwidth overload by excessive network connections, or by data overload; however, this is not distinguished by the method.
- The MITRE ATT&CK framework provides specific sub-techniques under each technique; however, some of them are incomplete, inconsistent, and narrowly defined.

IV. DIAMOND MODEL

The Diamond model utilises the diamond structure to emphasize the relationships and characteristics of an attack based on its four core components: adversary, infrastructure, capability, and victim [6]. This model explains how an *adversary* exploits a *capability* over an *infrastructure* against a *victim* (see Fig. 3) [17]. These four main components of an attack are the vertices of the diamond that gives this model its name. It further defines additional meta-features to support higher-level constructs and applies measurement, testability, and repeatability to provide a more comprehensive scientific method of analysis. This model was released by the US Department of Defense in 2013 [6]. The diamond model is a cognitive model in addition to a set of mathematical techniques [17]. The cognitive model allows security experts to organize large amounts of interrelated logic, whereas a set of mathematical techniques enables them to enhance strategic decision-making and analytical workflow against the adversary [17].

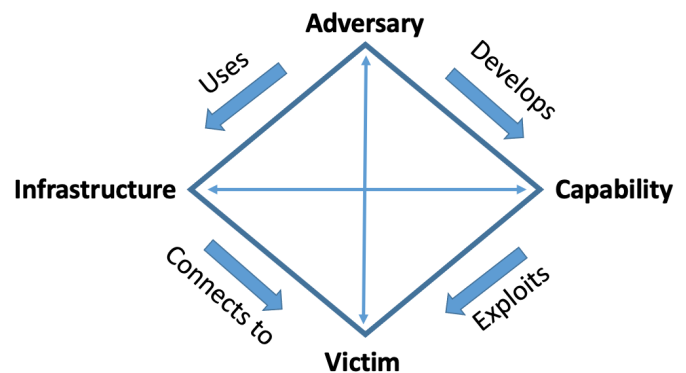


Fig. 3. Diamond Model of Intrusion Analysis

TABLE I
COMPARATIVE STUDY OF ATTACK MODELS: LOCKHEED MARTIN'S CYBER KILL CHAIN, MITRE ATT&CK FRAMEWORK AND DIAMOND MODEL

Criteria	Lockheed Martin's Cyber Kill Chain	MITRE ATT&CK Framework	Diamond Model
Year	2011	2013	2013
Creator	Lockheed Martin	MITRE	US Department of Defence
Key Feature	It provides a sequence of steps of an attack life cycle and on that basis, it aids development of a mechanism for interrupting any step of the attacker's kill chain as its possible defence.	It provides a knowledge base of adversarial tactics and techniques based on real-world observations, reflecting the various phases of an adversary's attack lifecycle and the platforms they are known to the target and their possible defence.	It provides a description of the relationships and characteristics of an attack based on four core features: adversary, infrastructure, capability, and victim. The model explains how an adversary exploits a capability over an infrastructure against a victim, to develop a possible defence.
Operational Structure	This model is a sequence of ordered steps following the anatomy of an attack including: Reconnaissance, Weaponization, Delivery, Exploitation, Installation, Command and Control, and Action on Objectives.	This model is a collection of tactics, consisting of an array of techniques and sub-techniques. The tactics are as follows: Initial Access, Execution, Persistence, Privilege Escalation, Defence Evasion, Credential Access, Discovery, Lateral Movement, Collection, Command & Control, Exfiltration, Impact	This model comprises a diamond structure where the four vertices of the diamond represent the four core features: Adversary, Infrastructure, Capability, and Victim.
Ease of Use	Complex	Complex	Complex
Mitigation	By interrupting any step of the cyber kill chain	By mapping attack patterns against courses of action	By treating the infrastructure, capability or victim
Application	Mainly aimed at large networked or enterprise systems	Mainly aimed at large networked or enterprise systems	Mainly aimed at large networked or enterprise systems

A. Components of Diamond Model

The Diamond Model mostly contains events, four core features, six meta features, and confidence values of features [6].

1) *Event*: An event is an atomic element of the Diamond Model. It defines a separate time-based activity limited to a certain phase where an adversary, necessitating external resources, uses a capability and methodology over some infrastructure against a victim with a given result.

2) *Core Features*: The Diamond Model consists four core features: adversary, capability, infrastructure, and victim as shown in Fig. 3. These core features are connected by edges on the Diamond structure, which define a distinctive relationship between each of the features. This relationship between features is based on analytic pivoting and the criteria on which a security expert can move from one point to another connected point on the Diamond structure.

Adversary: The adversary describes the actor who is responsible for utilising a capability against the victim to achieve an attack goal. Initially, the information about an adversary is difficult to find, therefore, the adversary may be empty for most events during the time of detection.

Capability: The capability describes the tools and techniques which are used in the event by the adversary to achieve an attack goal. The capability should include all methods to exploit the victim ranging from the most manual methods to the most sophisticated automated methods.

Infrastructure: The infrastructure describes the physical and/or logical communication structures which are used by the adversary to deliver and maintain control of capabilities and achieve an attack goal against the victim. The infrastructure can be classified in two categories: type 1 infrastructure, which is controlled by the adversary; and type 2 infrastructure, which is controlled by an intermediary.

Victim: The victim describes the target of the adversary and against whom vulnerabilities and exposures are exploited utilising the capabilities through the infrastructure. The victim can be considered anything which is pertinent to an event such as organization, person, email address, IP address, or domain name.

3) *Meta Features*: The Diamond Model also outlines several extra meta-features apart from the four core features. These meta features are: timestamp, phase, result, direction, methodology, resources. Furthermore, each core feature and meta feature can be expanded and may be composed of many sub-features, which in themselves may be composed of sub-features allowing the model to become as complex and rich as is necessary to fully describe the activity.

4) *Confidence Value*: Each event feature is assigned a confidence value, it is either a core feature or meta-feature. The confidence value is utilised in the analysis, and it could be a function of multiple values, such as the confidence of an analytic conclusion and/or the accuracy of a data source. The confidence value is decided dependent on the particular

implementation of the model which could be different for different implementation.

B. Advantages of Diamond Model

The Diamond Model has the following advantages [6], [17]:

- The diamond model incorporates scientific principles of hypothesis generation, correlation, measurement, testability, and repeatability to intrusion analysis with the intention of improving analytic effectiveness, efficiency, and accuracy.
- The diamond model enables the integration of cognitive intelligence in real-time for cyber analysis of an attack and its defence, automating correlation across events, classifying events with confidence into attacker campaigns, and forecasting attacker operations while planning mitigation strategies.
- The diamond model strengthens and advances the capabilities of organizations for performing cyber analysis by expanding and encompassing new ideas and concepts and formalizing first principles upon which new concepts can be explored.
- The diamond model enables the development of new and more effective mitigation strategies that can increase the cost on the attacker while reducing the cost to the defender.
- The diamond model enables contextual and relationship-rich indicators, thus improving cyber threat intelligence sharing and increasing the range of applicability of indicators.

C. Disadvantages of Diamond Model

The Diamond Model has the following disadvantages [3], [5], [6], [18]:

- The diamond model has a simple appearance, but it is complex to establish a mathematical or statistical framework upon which to apply advanced concepts such as clustering, classification, game theory, graph analysis, and other methods.
- Furthermore, the diamond model defines several additional meta-features apart from the four core features, where each core and meta-feature can be expanded and composed of many sub-features, which in themselves may be composed of sub-features, thus making the diamond model very complex.
- The diamond of a threat actor is not static but is in constant flux as attackers alter their infrastructure and/or capabilities frequently. An attacker's capabilities require significant effort to build and are relatively static, however, an attacker's infrastructure can be changed easily, which can be used to link different attack campaigns together and to a particular attacker. This infrastructure replacement can lead to higher false positives as several different actors can use the same infrastructure, meaning that as an attribution tool, it is not perfect due to the cognitive and mathematical techniques involved in the model.

- The diamond model utilises both cognitive and mathematical techniques, therefore, to maximise the efficiency of this technique, security experts are required to be highly skilled in these techniques.

V. COMPARATIVE STUDY OF ATTACK MODELS: LOCKHEED MARTIN'S CYBER KILL CHAIN, MITRE ATT&CK FRAMEWORK AND DIAMOND MODEL

This section presents a comparative study of the three previously examined attack models: the Lockheed Martin's Cyber Kill Chain, the MITRE ATT&CK Framework and the Diamond Model. This comparative study is shown in Table I, which covers a relative summary of these three attack models for use in selecting the most suitable attack model for a specific application.

VI. CONCLUSION

This paper presented a comparative study of the three popular attack models: the Lockheed Martin Cyber Kill Chain, the MITRE ATT&CK Framework and the Diamond Model. Attack models are very useful tools to analyse the cyber-attack lifecycle used by attackers to mount their attacks. All the models offer different characteristics, advantages and disadvantages; therefore, their suitability may differ from one attack analysis to another. This paper examined characteristics, advantages and disadvantages of these three attack models, and presented their comparative summary for use in selecting the most suitable attack model for a specific application. This comparative analysis provided an understanding of three popular attack models; however, these are not the only models which can be utilised for attack analysis. There are several other attack models which can also be utilised for attack analysis; therefore, it requires further comparative analysis encompassing a wider selection of attack models to offer a broader view and understanding in the area of attack analysis.

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